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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/819,176	08/11/2022	Takuya TANAKA	740165-F03509	8086

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EXAMINER
SMITH, ANNA ELIZABETH

ART UNIT	PAPER NUMBER
2872	

NOTIFICATION DATE	DELIVERY MODE
01/24/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

info@sbpatentlaw.com

Office Action Summary	Application No. 17/819,176	Applicant(s) TANAKA et al.	
	Examiner ANNA SMITH	Art Unit 2872	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 08/11/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☑ Claim(s) 2 and 18 is/are allowed.

7) ☑ Claim(s) 1,3,5-7,9-10,12,14,16-17 and 20 is/are rejected.

8) ☑ Claim(s) 4,8,11,13,15 and 19 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 08/11/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

2. Receipt is acknowledged of certified copies of papers required by 37 CFR 1.55.

Information Disclosure Statement

3. The information disclosure statement (IDS) submitted on 08/11/2022 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Claim Rejections - 35 USC § 103

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claims 1, 3, 5-7, 9, 10, 12, 14, 16-17 and 20 is/are rejected under 35 U.S.C. 103 as being unpatentable over Maetaki (US Patent Publication 20200012073 - cited by applicant).

6. **Regarding claim 1**: Maetaki teaches (embodiment 1, fig. 1) an imaging lens consisting of, in order from an object side to an image side: a first lens group that has a positive refractive power (first lens unit L1; positive e refractive power in the tables under paragraph [0101]), a second lens group that has a positive refractive power (second lens unit L2 – positive refractive power in the tables under [0101]), a third lens group (third lens unit L3), wherein: during focusing the second lens group moves along an optical axis while the first

and third lens group remain stationary (See fig. 1 where the movement of the imaging lens is restricted to the second lens unit), the second lens group includes a stop (fig. 1 SP), an Lp1 lens which is a positive lens (the 3rd lens is positive), and is disposed closest to the object side in the second lens group (see fig. 1), an Lp2 lens, which is a positive lens (the 9th lens is positive), and is disposed closest to the image side of the second lens group (See fig. 1) and wherein the following conditional expression is satisfied: 28.4<vn<40 (from the tables under [0101] vn = 32.26).

7. Maetaki additionally teaches wherein the average value of the refractive index of Lp1 and Lp2 is Np12 = 1.763, which closely approaches the claimed range of 1.94<Np12<2.5. However, the value disclosed by Maetaki is so close to the claimed range that one of ordinary skill in the art would expect the imaging lens to have the same properties. See *Titanium Metals Corp. v. Banner,* 778 F.2d 775, 227 USPQ 773 (Fed. Cir. 1985) and MPEP 2144.05. Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the invention to have the imaging lens satisfy the conditional expression for the purposes of favorably correcting optical aberrations (Maetaki [0073]), and since the disclosed value is so close to the claimed range that one of ordinary skill int eh art would expect the imaging lens to have the same properties.

8. **Regarding claim 3**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein the following conditional expression is satisfied: 0.5<D3/BF<1 (from the tables below [0101] D3/BF = 1.83).

9. **Regarding claim 5**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein the following conditional expression is satisfied: 0.1<f/f1<0.3 (from the tables below [0101] f/f1=0.191).

10. **Regarding claim 6**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein the second lens group includes a cemented lens in which Ln2 lens as a negative lens and the Lp2 lens are cemented in order from object side (the 8th and 9th lens are cemented together, where the 8th lens has a negative power – see tables).

11. **Regarding claim 7**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein the following conditional expression is satisfied: 0.3<Np2-Nn2<0.7 (from the tables below [0101] Np2-Nn2 = 0.35).

12. **Regarding claim 9**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein the Lp1 lens is a positive meniscus lens that has a concave surface facing towards the image side (see figure 1).

13. **Regarding claim 10**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein the second lens group includes at least two positive lens (lenses 3 and 4) and one negative lens (5th lens) at a position closer to the object side than the stop (these three lenses are on the left side of the stop); and includes at least two positive lens (lenses 7 and 9) and two negative lenses (lenses 6 and 8) at a position closer to the image side than the stop (theses lenses are on the right side of the stop closer to the image side).

14. **Regarding claim 12**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein the following conditional expression is satisfied: 0.5<BF/f*tan(wm)<1 (from the tables below [0101] BF/ftan(wm)=0.689).

15. **Regarding claim 14**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki discloses wherein f1/BF = 30.3, which closely approaches the claimed range of 20<f1/BF<30. However, the value disclosed by Maetaki is so close to the claimed range that one of ordinary skill in the art would expect the imaging lens to have the same

properties. See *Titanium Metals Corp. v. Banner,* 778 F.2d 775, 227 USPQ 773 (Fed. Cir. 1985) and MPEP 2144.05. Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the invention to have the imaging lens satisfy the conditional expression for the purposes of favorably correcting optical aberrations (Maetaki [0073]), and since the disclosed value is so close to the claimed range that one of ordinary skill int eh art would expect the imaging lens to have the same properties.

16. **Regarding claim 16**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki discloses wherein the average value of the refractive index of Lp1 and Lp2 is Np12 = 1.763, which closely approaches the claimed range of 1.965<Np12<2.2. However, the value disclosed by Maetaki is so close to the claimed range that one of ordinary skill in the art would expect the imaging lens to have the same properties. See *Titanium Metals Corp. v. Banner,* 778 F.2d 775, 227 USPQ 773 (Fed. Cir. 1985) and MPEP 2144.05. Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the invention to have the imaging lens satisfy the conditional expression for the purposes of favorably correcting optical aberrations (Maetaki [0073]), and since the disclosed value is so close to the claimed range that one of ordinary skill int eh art would expect the imaging lens to have the same properties.

17. **Regarding claim 17**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein the conditional expression 29<vn<35 is satisfied (Maetaki discloses vn=32.26).

18. **Regarding claim 20**: Maetaki teaches the imaging lens according to claim 1, as set forth above. Maetaki further teaches wherein an imaging apparatus comprises the imaging lens according to claim 1 (See fig. 13).

Allowable Subject Matter

19. Claims 4, 8, 11, 13, 15 and 19 objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

20. The prior art taken singularly or in combination fail to anticipate or suggest all of the limitations of claims 4, 8, 11, 13, 15 and 19 including, with all other limitations:

21. With regards to claim 4, wherein the conditional expression 3<D2/D3<5 (Maetaki discloses the expression to be 2.07) is met;

22. With regards to claim 8, wherein the conditional expression 5<vn2-vp2<30 (Maetaki discloses the expression to be -11.5) is met;

23. With regards to claim 11, wherein the third lens group consists of a cemented lens with a positive lens and a negative lens and wherein the negative lens has a concave surface facing towards the object side;

24. With regards to claim 13, wherein the conditional expression $7.1 < \frac{TL^2}{f^2 \tan(wm)} < 11$ (Maetaki discloses the expression to be 11.92) is met;

25. With regards to claim 15, wherein the conditional expression 1.2<D12/(ftan(wm))<3 (Maetaki discloses the expression to be 0.66) is met;

26. With regards to claim 19, wherein the conditional expression $7 < \frac{TL^2}{f^2 \tan(wm)} < 10$ (Maetaki discloses the expression to be 11.92) is met, in such a manner where a rejection under 35 U.S.C. §102 or §103 would be appropriate.

27. Claims 2 and 18 allowed.

28. The following is a statement of reasons for the indication of allowable subject matter: The prior art taken singularly or in combination fail to anticipate or suggest all of the limitations of claims 2 and 18 including:

29. With regards to claim 2, wherein an imaging lens consisting of a first lens group that has a positive refractive power; a second lens group that has a positive refractive power; and a third lens group, an Lp1 lens, which is a positive lens, is disposed closest to the object side in the second lens group, an Lp2 lens, which is a positive lens, is disposed closest to the image side in the second lens group, and wherein the conditional expression 0.5<D3/BF<1 (Maetaki discloses the expression to be 1.83) is met;

30. With regards to claim 18, for being dependent on claim 2 and wherein the following conditional expression is satisfied 0.7<D3/BF<1 (Maetaki discloses 1.83), in such a manner where a rejection under 35 U.S.C. §102 or §103 would be appropriate.

Conclusion

31. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

32. Maruyama (US Patent Publication 20180372986), Iwamoto (US Patent Publication 20200341248), Eguchi (US Patent Publication 20200348496) and Ohtake (US Patent Publication 20240201475) all disclose imaging lenses that are similar to the claimed invention, but do not meet all of the limitations of the independent claims.

33. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANNA SMITH whose telephone number is (571)270-1401. The examiner can normally be reached Mon-Fri 8:00 am - 4 pm (MST).

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ricky Mack can be reached on (571) 272-3333. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ANNA SMITH/
Examiner, Art Unit 2872
1/15/2025

/George G. King/
Primary Examiner, Art Unit 2872